EWB COURSES

AI-POWERED CYBERSECURITY

Assessment 01: Security Foundations

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Batch/Group: 1

Course: AI-Powered Cybersecurity

Assessment: Individual Work

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Tasks Covered:

- Task A: Secure a Student Account
- Task B: Analyse a Suspicious Message
- Task C: Triage a Sign-in Alert
- Task D: Review an AI Security Answer

TASK A: SECURE STUDENTS ACCOUNT

A1. Identify Five Security Risks

Mira's situation has five main security risks that can affect her accounts, laptop and project files.

Risk	Asset	CIA Impact	Control	Limitation
Reusing a short password	**Email and learning portal**	**Confidentiality may be affected if both accounts are accessed.**	**Use a strong, unique password for each account.**	**Passwords can still be stolen through phishing.**
Approving unexpected login prompts	**Student account**	**Unauthorised access may expose or change information.**	**Deny unknown prompts and report repeated requests.**	**It depends on the user's response.**
Delaying device updates	**Laptop and files**	**Known weaknesses may affect confidentiality, integrity or availability.**	**Install security updates regularly.**	**Updates cannot prevent every attack.**
Keeping only one project copy	**Project files**	**Availability is affected if files are lost.**	**Keep a separate backup.**	**Backups may be outdated or damaged.**
Sharing a signed-in laptop	**Private information and assignments**	**Another person may view or change files.**	**Lock the laptop and sign out before sharing.**	**Weak security may still allow access.**

A2. Practical Controls and Priorities

I would take the following actions :

Priority 1: Mira should deny unexpected login prompts and report repeated requests. This is important because approving an unknown prompt may allow someone to access her account. However, it depends on her recognising suspicious activity.

Priority 2: She should change her reused password and create unique passwords for both accounts. This reduces the risk of one exposed password affecting multiple accounts. However, even a strong password cannot prevent every phishing attack

Priority 3: She should lock her laptop, sign out before sharing it and install pending updates. These actions reduce unauthorised access and protect against known software weaknesses.

After these steps, Mira should back up her project files to a separate, secure location and check that the backup works. She should review these security practices regularly.

A3. Authentication and Authorisation

Authentication means verifying a user's identity. For example, Mira enters her username and password and completes the required multi-factor authentication to access her learning portal.

Authorisation means deciding what an authenticated user is allowed to do. For example, Mira can view her own marks and submit assignments, but she cannot change another student's marks.

In simple words, authentication checks who the user is, while authorisation checks what the user is allowed to access.

Task B: Analyse a Suspicious Message

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B1. Identify Five Warning Signs

Warning sign from message	Explanation
“FINAL WARNING”	This creates fear and pressure on the student to take quick action without checking properly.
“expires in 20 minutes”	The short deadline creates urgency and may stop the student from verifying the scholarship.
“Reply-To: help@personal-awards.example”	The reply address is different from the normal college domain, so it needs to be checked.
“enter your college password and OTP”	Asking for a password and OTP is risky because someone may use these details to access the student's account.
“Do not contact the college office”	This prevents the student from checking whether the scholarship is real. The request for an identity card photo is also a concern because it contains personal information

B2. Safe Verification Process

1. I would not click the link or reply to the message.

2. I would open the official college portal using a saved bookmark or an address obtained independently.

3. I would contact the college office or scholarship department using the official contact details to confirm the message.

4. I would report the suspicious message to the college IT or security team.HTTPS only means that the connection between the browser and website is encrypted. A fake website can also use HTTPS. Also, a familiar sender name can be copied, so it does not prove that the message is genuine.

B3. Response Plan if Password and OTP Were Entered

1. Immediate action: I would contact the authorised IT team and change the password through the official portal using a trusted device.

2. Secure the account: I would ask IT to revoke active sessions and check MFA settings, recovery details and unknown devices.

3. Report the incident: I would report the message to the college security team and explain what information was entered. I would not share the actual password or OTP.

4. Follow-up checks: I would review recent login activity and check for unknown changes. If I used the same password on other accounts, I would change those passwords too.

B4. Limit of Evidence

From this message alone, I cannot confirm who actually sent it or whether the website is malicious. Verified email headers, sender-domain information and an authorised investigation of the website would help check its authenticity. Until then, I would treat the message as suspicious and avoid sharing any information.

TASK C: TRIAGE A SIGN-IN ALERT

Task C: Triage a Sign-in Alert

The given sign-in log shows some unusual activity on the student account. There were two failed password attempts followed by a successful password login from a different IP address. MFA was denied first and then approved. This activity needs further investigation before deciding what happened.

C1. Four-Point Timeline

Time	Observed event	Interpretation
09:00–09:01	Password was successful and MFA was approved from 192.0.2.10.	This appears to be a normal sign-in from the usual classroom network.
10:14–10:15	Two password attempts failed from 198.51.100.23.	Someone may have tried to access the account, but the reason is not confirmed.
10:16–10:18	Password succeeded, MFA was denied and then approved. The student said they approved a prompt they did not request.	This raises concern that someone may have known the password and the approval allowed the login to continue.
10:20	A session was created from 198.51.100.23.	A session was established after authentication. IT should check whether it was authorised and what activity followed.

C2. Possible Explanations and Additional Checks

Likely explanation: Someone may have obtained the student's password and tried to sign in. The student approved the unexpected MFA prompt, which may have allowed the login to continue.

Alternative explanation: The login may have been initiated by the student or an authorised service using a different network, although the student's statement makes this less likely.

Additional checks:

1. Review device, browser, timestamp, MFA and session activity details to check whether the login matches the student's normal usage.

2. Contact the student through an authorised channel and confirm their activity. Compare the information with available records. The IP address alone cannot identify the person.

C3. Three Prioritised Response Actions

1. Protect the account: Reset the password through the approved process, revoke active sessions and review MFA settings. Check recovery details and unknown devices.

2. Preserve evidence: Save the relevant authentication, MFA and session logs with timestamps before they are removed by routine retention. Only authorised staff should access them.

3. Investigate the activity: Review what happened after the session was created, check for unauthorised changes and document the findings and any remaining uncertainty.

C4. Incident Summary

The log shows a normal sign-in at 09:00–09:01, followed by two failed password attempts and a successful login from a different IP at 10:16. MFA was denied and then approved, and a session was created at 10:20. The student says they did not initiate the login but approved the prompt. This may indicate unauthorised access, but the evidence does not identify a person or prove how the password was obtained. IT should secure the account, preserve the logs and investigate further..

TASK D: REVIEW AN AI SECURITY ANSWER

D1. Identify and Correct Five Unsafe Claims

AI statement or action	Correction and reason
"The email is safe because it uses HTTPS."	HTTPS encrypts the connection between the browser and website, but phishing websites can also use HTTPS. The sender and website must be checked independently.
"The logs prove that the person using 198.51.100.23 stole the password."	The logs show sign-in events from an IP address, but they do not identify the person or prove how the password was obtained. More evidence is required.
"MFA prevents every account compromise."	MFA adds an extra layer of security but does not prevent every attack. In this case, the student approved an unexpected MFA prompt, which may have allowed the login to continue.
"Delete the suspicious logs."	Relevant logs should be preserved because they help the IT team investigate the incident. Deleting them may remove important evidence.
"Send the student's password to support so they can test it."	Passwords and OTPs should never be shared with support staff. The IT team should use approved account recovery and verification procedures

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D2. Improved Prompt for an AI Tool

Analyse only the fictional email and synthetic sign-in logs provided in this assessment. Identify possible security risks and recommend safe response actions. Clearly separate observed facts from assumptions. Do not identify a person based only on an IP address. Do not request or reveal passwords, OTPs or other private information. State what cannot be confirmed from the available evidence and suggest additional checks that an authorised IT team can perform.

D3. Corrected Security Advice

Do not trust an email only because it uses HTTPS or has a familiar sender name. Verify the message through the official college portal or contact the college office using independently obtained contact details. Never enter your password or OTP on an unverified website. In the given sign-in log, two failed password attempts were followed by a successful login, an MFA denial and approval, and a new session. This activity is suspicious, but the IP address does

not prove who was responsible. If you approved an unexpected prompt or entered your login details, contact the authorised IT team immediately. Change your password through the official portal, revoke unknown sessions and check MFA settings. The IT team should preserve relevant logs and investigate the session activity before reaching a final conclusion.

D4. AI-Use Declaration and References

AI-use declaration: ChatGPT was used to help organise the draft answers, explain cybersecurity concepts and improve clarity. The final answers should be reviewed and edited to reflect my own understanding.

Evidence and reference notes:

1. CISA – Secure Our World: https://www.cisa.gov/secure-our-world

This reference provides basic guidance on strong passwords, multi-factor authentication and recognising or reporting phishing attempts.

2. NIST – Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (NIST AI 600-1): https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

This reference explains risks associated with AI-generated information and the importance of checking AI outputs instead of accepting them without verification.